THE COHERENT MODULES FORM AN ABELIAN CATEGORY AND ARE STABLE UNDER KERNELS, COKERNELS AND EXTENSIONS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The two clauses of the definition of a coherent module: local generation by finitely many sections and local finite generation of the kernels of the induced maps. Invoked in the counted proof (source lines 1881-1892) as a pointer to the definition of this chapter. Supplies both the statement that a subsheaf inherits and the finiteness input for the kernel of a map into a coherent sheaf.
- The finiteness of extensions of modules of finite type, and the criterion for a category to be a weak Serre subcategory. Invoked inside the counted proof (source lines 1948 and 1979-1980) as a pointer to a lemma of this chapter and to a result of another chapter. Gives finite typeness of the middle term of a short exact sequence and the last assertion.

Source reference. Original definition definition-coherent: sources/stacks-modules/pinned-original-source.tex, source lines 1881–1881. Navigation pointer only: the counted text refers to the Stacks item definition-coherent, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-extension-finite-type: sources/stacks-modules/pinned-original-source.tex, source lines 1667–1667. Navigation pointer only: the counted text refers to the Stacks item lemma-extension-finite-type, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

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Source reference. Original reference homology-lemma-characterize-weak-serre-subcategory: sources/stacks-modules/pinned-original-source.tex, source lines 1980–1980. Navigation pointer only: the counted text refers to the Stacks item homology-lemma-characterize-weak-serre-subcategory, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space.*

(1) *Any finite type subsheaf of a coherent sheaf is coherent.*
(2) *Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism from a finite type sheaf $\mathcal{F}$ to a coherent sheaf $\mathcal{G}$. Then $\mathrm{Ker}(\varphi)$ is of finite type.*
(3) *Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of coherent $\mathcal{O}_X$-modules. Then $\mathrm{Ker}(\varphi)$ and $\mathrm{Coker}(\varphi)$ are coherent.*
(4) *Given a short exact sequence of $\mathcal{O}_X$-modules $0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0$ if two out of three are coherent so is the third.*
(5) *The category $Coh(\mathcal{O}_X)$ is a weak Serre subcategory of $Mod(\mathcal{O}_X)$. In particular, the category of coherent modules is abelian and the inclusion functor $Coh(\mathcal{O}_X) \to Mod(\mathcal{O}_X)$ is exact.*

Proof. Condition (2) of Definition definition-coherent holds for any subsheaf of a coherent sheaf. Thus we get (1).

Assume the hypotheses of (2). Let us show that $\mathrm{Ker}(\varphi)$ is of finite type. Pick $x \in X$. Choose an open neighbourhood U of x in X such that $\mathcal{F}|_U$ is generated by $s_1, \ldots, s_n$. By Definition definition-coherent the kernel $\mathcal{K}$ of the induced map $\bigoplus_{i=1}^n \mathcal{O}_U \to \mathcal{G}$, $e_i \mapsto \varphi(s_i)$ is of finite type. Hence $\mathrm{Ker}(\varphi)$ which is the image of the composition $\mathcal{K} \to \bigoplus_{i=1}^n \mathcal{O}_U \to \mathcal{F}$ is of finite type.

Assume the hypotheses of (3). By (2) the kernel of φ is of finite type and hence by (1) it is coherent.

With the same hypotheses let us show that $\mathrm{Coker}(\varphi)$ is coherent. Since $\mathcal{G}$ is of finite type so is $\mathrm{Coker}(\varphi)$. Let $U \subset X$ be open and let $\overline{s}_i \in \mathrm{Coker}(\varphi)(U)$, $i = 1, \ldots, n$ be sections. We have to show that the kernel of the associated morphism $\overline{\Psi} : \bigoplus_{i=1}^n \mathcal{O}_U \to \mathrm{Coker}(\varphi)$ is of finite type. There exists an open covering of U such that on each open all the sections $\overline{s}_i$ lift to sections s_i of $\mathcal{G}$. Hence we may assume this is the case over U. We may in addition assume there are sections t_j, $j = 1, \ldots, m$ of $\mathrm{Im}(\varphi)$ over U which generate $\mathrm{Im}(\varphi)$ over U. Let $\Phi : \bigoplus_{j=1}^m \mathcal{O}_U \to \mathrm{Im}(\varphi)$ be defined using t_j and $\Psi : \bigoplus_{j=1}^m \mathcal{O}_U \oplus \bigoplus_{i=1}^n \mathcal{O}_U \to \mathcal{G}$ using t_j and s_i. Consider the following commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \bigoplus_{j=1}^m \mathcal{O}_U & \longrightarrow & \bigoplus_{j=1}^m \mathcal{O}_U \oplus \bigoplus_{i=1}^n \mathcal{O}_U & \longrightarrow & \bigoplus_{i=1}^n \mathcal{O}_U & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \Phi} & & \downarrow{\scriptstyle \Psi} & & \downarrow{\scriptstyle \overline{\Psi}} & & \\ 0 & \longrightarrow & \mathrm{Im}(\varphi) & \longrightarrow & \mathcal{G} & \longrightarrow & \mathrm{Coker}(\varphi) & \longrightarrow & 0 \end{array}$$

By the snake lemma we get an exact sequence $\mathrm{Ker}(\Psi) \to \mathrm{Ker}(\overline{\Psi}) \to 0$. Since $\mathrm{Ker}(\Psi)$ is a finite type module, we see that $\mathrm{Ker}(\overline{\Psi})$ has finite type.

Proof of part (4). Let $0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0$ be a short exact sequence of $\mathcal{O}_X$-modules. By part (3) it suffices to prove that if $\mathcal{F}_1$ and $\mathcal{F}_3$ are coherent so is $\mathcal{F}_2$. By Lemma lemma-extension-finite-type we see that $\mathcal{F}_2$ has finite type. Let $s_1, \ldots, s_n$ be finitely many local sections of $\mathcal{F}_2$ defined over a common open U of

X. We have to show that the module of relations $\mathcal{K}$ between them is of finite type. Consider the following commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & 0 & \longrightarrow & \bigoplus_{i=1}^n \mathcal{O}_U & \longrightarrow & \bigoplus_{i=1}^n \mathcal{O}_U & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathcal{F}_1 & \longrightarrow & \mathcal{F}_2 & \longrightarrow & \mathcal{F}_3 & \longrightarrow & 0 \end{array}$$

with obvious notation. By the snake lemma we get a short exact sequence $0 \to \mathcal{K} \to \mathcal{K}_3 \to \mathcal{F}_1$ where $\mathcal{K}_3$ is the module of relations among the images of the sections s_i in $\mathcal{F}_3$. Since $\mathcal{F}_1$ is coherent we see that $\mathcal{K}$ is the kernel of a map from a finite type module to a coherent module and hence finite type by (2).

Proof of (5). This follows because (3) and (4) show that Homology, Lemma homology-lemma-characterize-weak-serre-subcategory applies. $\square$